Subject:	Re: Jamovi module - genetic epidemiology analysis
Date:	Monday, 27 April 2026 at 03:38:59 Central European Summer Time
From:	The jamovi Project
To:	MORENO AGUADO, VICTOR RAUL
Attachments:	Screenshot 2026-04-27 at 11.12.08.png, sample1.omv

Hi Victor,

A few suggestions.

Attached is an error I received while mucking around with it.

Your example data files are .rda files ... which jamovi can't actually open ... (if I navigate to the jamovi library, and select the examples provided, they won't open).

When I change an unrelated option, I notice that the SNP summary table disappears and reappears (see the following in slow mo):

https://youtu.be/UG3Gil5z0Hc

We try and avoid this. We try and keep the results as stable as possible as the user makes changes. If you use data binding, you can have jamovi set these tables up for you automatically:

https://dev.jamovi.org/tuts0201-dynamic-tables.html

This handles the simple condition where the user has "Stratify by response groups" turned off, and the table has one row per SNP var.

When you have Stratify by response groups turned on, I see that it becomes five rows per SNP variable. The way to accomodate this is to use "row folding" where you strategically name columns so that they fold down underneath one another.

Take a look at the independent samples t-test in jmv

Independent Samples T-Test

Independent Samples T-Test

		Statistic	df	p
len	Student's t	1.915	58.00	.0604
	Welch's t	1.915	55.31	.0606
	Mann-Whitney U	324.5		.0645
dose	Student's t	0.000	58.00	1.0000
	Welch's t	0.000	58.00	1.0000
	Mann-Whitney U	450.0		1.0000

Note. H_a $\mu_{OJ} \neq \mu_{VC}$

The use of square brackets in the column names causes the rows to fold into three rows

https://github.com/jamovi/jmv/blob/main/jamovi/ttestis.r.yaml#L20-L327

Lastly, if you use clearWith appropriately, you can prevent unrelated option changes from blanking a table, and then repopulating them with exactly the same values.

https://dev.jamovi.org/tuts0203-state.html

You can also populate the Alleles (A/B) column by implementing a .init() function alongside .run().

We're currently improving our documentation around this:

https://jamovi-dev.b-cdn.net/tutorial/tuts0200-analysis-lifecycle

Here's some LLM feedback:

1. Formula injection via unsanitized column names
Files: snpAssoc.b.R:16-21, snpLDHaplo.b.R:402
Severity: High (depends on jamovi's own column-name sanitization for mitigation)

Column names from the user's dataset are string-interpolated directly into R formula strings and then passed to as.formula():

```
cov_formula <- paste("+", paste(names(covariates_df), collapse = "+"))
formula_full <- as.formula(paste("resp ~ snp", cov_formula))
```

In R, formulas are evaluated as language objects. When lm() / glm() / haplo.glm() processes the formula, any R expression embedded in the column name string (e.g. a column literally named system("rm -rf /")) would be executed during model.frame() evaluation. This pattern appears in at least 12 places across the codebase.

JONATHON ADDS there's also composeFormula described here:

https://dev.jamovi.org/tuts0104-implementing-an-analysis.html

3. source() called at package load time
Files: snpAssoc.b.R:4, snpDesc.b.R:4, snpLDHaplo.b.R:6
Severity: Medium

Each .b.R file calls source("R/snp_helpers.R") at the top level (outside any function), so it runs when the package is loaded, not lazily. This is wrong for R packages:

Functions in snp_helpers.R should be in the package namespace (listed in NAMESPACE) — using source() bypasses that.
The relative path "R/snp_helpers.R" resolves against the process working directory, which is not guaranteed to be the package root in all contexts (tests, R CMD check, etc.).
Reloading the package in an interactive session would re-execute source() unexpectedly.
Fix: Remove all three source() calls. The functions in snp_helpers.R are already part of the package and will be available in the package namespace automatically

File: snpLDHaplo.b.R:597
Severity: Low

.compute_haplo_interaction_ (with a trailing underscore, ~270 lines) is never called. The actual live function is .compute_haplo_interaction defined at line 878. This dead code is a maintenance hazard and creates confusion about which implementation is current.

Fix: Delete the dead .compute_haplo_interaction_ method.

5. <<- inside lapply for side-effect tracking
File: snpAssoc.b.R:138-141

```
first_inter_done <- FALSE
result <- lapply(all_keep, function(r) {
...
if (attach_p) first_inter_done <<- TRUE # mutates parent env from inside lapply
...
})
```

This is fragile: it relies on <<- escaping to the enclosing function scope from inside a lapply callback. If the code is ever refactored (e.g., the lapply is extracted), this will silently break. It also makes the code hard to reason about.

Fix: Replace with a for loop and a regular <- assignment, or refactor to track the state through the return value.

6. Unhandled "categorical" response type

File: snp_helpers.R:272

else if (n_unique > 2 & n_unique <= 6) "categorical"

detect_response_type() can return "categorical" when the response has 3–6 unique values:

But all downstream model-fitting code (fit_model, fit_interaction_model, etc.) only branches on "binary" and "quantitative". When response_type == "categorical", the code falls through to the else branch silently, running a linear model (lm) on what the user likely intends to be a categorical outcome, producing nonsensical results without any warning.

Fix: Either explicitly handle "categorical" with a user-facing warning/error, or document that it maps to quantitative treatment (and change the detection logic to reflect that).

Cheers

jonathon

From: The jamovi Project <contact@jamovi.org>
Sent: 24 April 2026 12:38
To: MORENO AGUADO, VICTOR RAUL <v.moreno@iconcologia.net>
Subject: Re: Jamovi module - genetic epidemiology analysis

Hi Victor,

Great to hear from you. Yes, we'll take a look and come back to you.

Cheers

Jonathon

From: MORENO AGUADO, VICTOR RAUL <v.moreno@iconcologia.net>
Sent: 21 April 2026 00:19
To: The jamovi Project <contact@jamovi.org>
Subject: Jamovi module - genetic epidemiology analysis

Hi,
I have built a jamovi module specific for genetic epidemiology analysis.
This is a port (thanks to Claude) of a web we developed in 2006 (yes 20 years ago) that has been very successful: https://www.snpstats.net

The paper we published has had more than 2400 citations
doi.org/10.1093/bioinformatics/btl268
I may write an update paper to advertise this module and will put a link to jamovi in the snpstats website.

The module is here:
https://github.com/victor-moreno/SNPstats-jamovi

I only have tested it in mac, I hope there are no issues on other SOs.

The module depends on haplo.stat, which is “archived" in CRAN because it has some problems with new checks they perform. Fortunately, the rstudio repo that jamovi uses installs a version from 2025 that works well, though I had to extract and include a couple of internal functions that, though exported, were not found by haplo.glm. This might be a problem in future updates.

I also have created a logo (in folder docs).

This is my first jamovi module and I don’t know how it works to send the module to the jamovi library.
In case I need to make any changes and improvements, once I push the changes, will they be propagated to the library?

For now it is only in English. I may add internationalization in the future.

Best regards

Victor